Notice, question, predict

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use observations to make a prediction that can be tested.

SUCCESS CRITERIA - I can...

1. Separate stated evidence from unsupported claims.
2. Extend a position pattern and state an assumption.
3. Name an observation that could test the prediction.

Core: teacher selects from 1-9. **Stretch:** 10 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Read a scale

A ruler shows a toy car at 2 cm, then at 7 cm. Draw both positions and the change. How far apart are the marks?

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $\Delta x = x_f - x_i$; $x_{next} = x_{now} + \Delta x$.
For a repeating position pattern only. i = initial; f = final. Check the pattern against evidence.

Worked reasoning: A cart is at 2, 5 and 8 m at 0, 1 and 2 s. Equal jumps of 3 m suggest 11 m at 3 s, if the pattern continues.

Your turn: A motion sensor recorded the first three positions in the table. Add the predicted position at 3 s. Draw the four positions on a line and distinguish the prediction from the measurements.

Time (s)	Position (m)
0	2
1	5
2	8
3	Predict: ____

Marzano check - circle: ① ② ③

03 What does the sensor show?

A class uses a motion sensor to track a cart. The table shows its position once each second. Describe two patterns and circle the measurements that support each one.

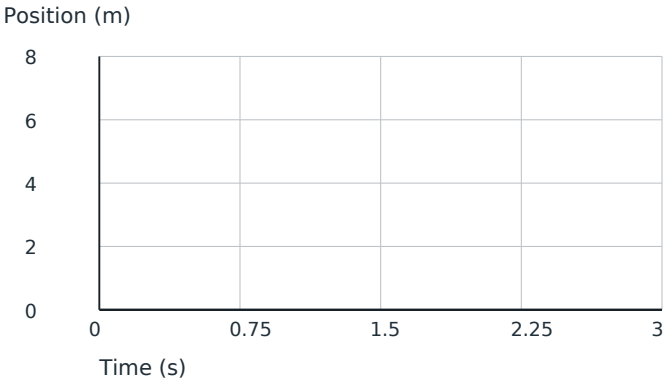
Support: I notice ____. The data show ____.

Time (s)	Position (m)
0	1
1	3
2	5

Marzano check - circle: ① ② ③

04 Make a prediction

Use the measurements in question 3 to predict where the cart will be at 3 s. Add the prediction to the blank graph. Show the pattern you used.



Marzano check - circle: ① ② ③

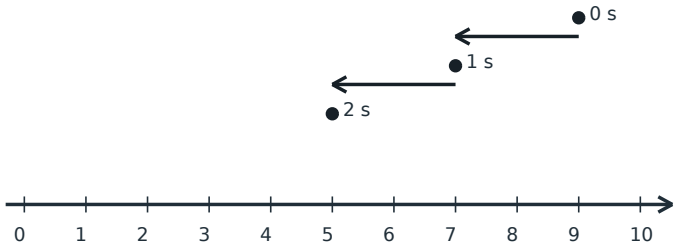
05 Evidence or a claim?

A fictional space report says: "An object is approaching Earth, but its future path is uncertain." For each statement, write **supported** or **not established**:
(a) It is getting closer.
(b) It will definitely hit Earth.
(c) More observations are needed to predict its path.

Marzano check - circle: ① ② ③

06 Test the model

A different cart is observed at 9, 7, 5 m at 0, 1, 2 s. Predict its position at 3 s. What would a new reading of 4 m make you check?



Position (m); right is positive

Marzano check - circle: ① ② ③

07 Ask a question you can investigate

A classmate asks, "Should we worry about the asteroid?" Rewrite this as a question that a measurement could help answer. Name the measurement and explain how it would help.

Support: I could investigate ____. I would measure ____.

Marzano check - circle: ① ② ③

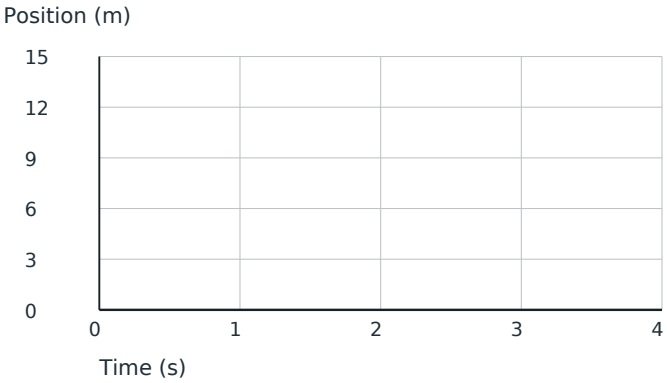
08 Choose useful evidence

Two groups time the same toy car over the same marked track. One records three times; the other writes "pretty fast." Which record would let someone check the claim? Explain.

Marzano check - circle: ① ② ③

09 Exit check

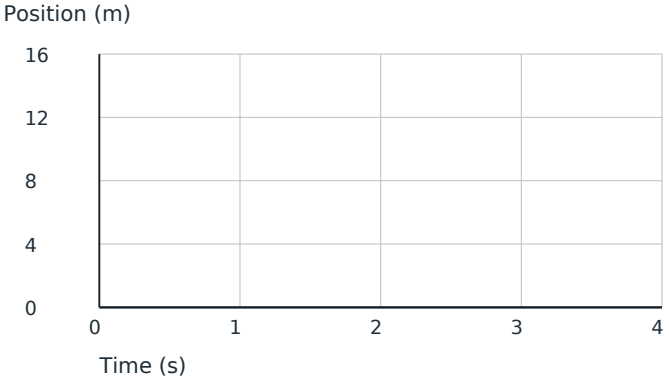
A robot moves along a marked floor. Its positions are 0, 3 and 6 m at 0, 1 and 2 s.
(a) Plot the measurements and predict its position at 4 s.
(b) State your assumption and one later observation that would challenge it.



Marzano check - circle: ① ② ③

10 STRETCH: test the model

Return to the worked example in problem 2. Create two possible positions at 4 s if the cart changes its motion after 2 s. Draw both possibilities. What measurement would help choose between them?



Marzano check - circle: ① ② ③